School of Innovation and Management (SoIM)
Post Graduate Diploma in Management (PGDM)

Course Outline **PGDM (2025-27)** **Trimester – IV**

Course Code: 404OM2	Course Name: Total Quality Management	Course Credits: 3

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COURSE COVERAGE:

This course provides a comprehensive grounding in quality management, diagnostic frameworks, and continuous improvement methodologies essential for operational excellence. Beginning with foundational quality leadership and historical evolution, learners explore customer-centric frameworks such as Voice of the Customer (VOC) and Kano Model, alongside measurement systems, key performance indicators (KPIs), and Cost of Quality (COQ) metrics.

The curriculum delves into core diagnostic tools, including Root Cause Analysis (5 Whys, Fishbone), Process Mapping, Value Stream Mapping (VSM), Failure Mode and Effects Analysis (FMEA), and statistical techniques (SPC, Control Charts, Pareto, Regression). The course concludes with strategic quality frameworks (QFD, Six Sigma, Lean, EFQM/Baldrige) and modern digital quality applications, equipping managers to build interactive dashboards and lead quality-driven transformations in Industry 4.0 environments.

COURSE OBJECTIVES:

By the end of this course, students will be able to:

1. Trace the evolution of quality management from inspection to Total Quality Management (TQM) and apply core leadership philosophies.
2. Diagnose customer requirements using the Kano Model, SERVQUAL, and Voice of the Customer (VOC) tools.
3. Establish robust quality measurement systems, key performance indicators (KPIs), and Cost of Quality (COQ) metrics.
4. Execute continuous improvement methodologies including PDCA, DMAIC, and Kaizen.
5. Apply root-cause analysis tools (5 Whys, Fishbone Diagram) to resolve business process failures.
6. Develop detailed Process Maps and Value Stream Maps (VSM) to identify and eliminate process waste.

7. Perform Failure Mode and Effects Analysis (FMEA) to mitigate potential process and product risks.
8. Construct and interpret Statistical Process Control (SPC) charts, Pareto diagrams, histograms, and run charts using software tools.
9. Apply statistical correlation and regression techniques to analyze and solve complex quality problems.
10. Deploy strategic alignment toolkits, including Quality Function Deployment (QFD), Lean Six Sigma, Benchmarking, and Business Excellence frameworks.
11. Formulate change management strategies to foster employee involvement and build a sustainable quality culture.
12. Leverage digital tools, IoT, AI, and visualization dashboards (Excel/Power BI) to drive smart manufacturing and strategic quality integration.

COURSE LEARNING OUTCOMES:

After completing the course, students will be able to demonstrate proficiency in:

CO1 - Quality Leadership & Foundations: Apply seminal quality philosophies (Deming, Juran, Crosby) and organizational behavior concepts to build a culture of continuous improvement and customer-centric quality.

CO2 - Measurement Systems & Problem Diagnostics: Quantify quality metrics, capture Voice of the Customer (VOC), evaluate Cost of Quality (COQ), and conduct risk analysis using FMEA, 5 Whys, and Fishbone techniques.

CO3 - Process Mapping & Statistical Analysis: Map end-to-end process workflows, construct Statistical Process Control (SPC) charts, and apply analytical statistical tools (Pareto, Histograms, Regression) to minimize defect rates.

CO4 - Strategic Quality & Digital Transformation: Integrate strategic frameworks (QFD, Six Sigma, Lean, EFQM/Baldrige), manage organizational change, and design digital quality dashboards (Excel/Power BI) for Industry 4.0 operations.

ASSESSMENT TOOLS FOR LEARNING OUTCOME:

Learning Outcome	Assessment Criteria
Evaluated via active class participation, case study analyses on quality leadership, and group activities.	20%
Assessed through problem-solving exercises on COQ, VOC workshop outputs, and FMEA activity submissions.	25%
Measured via hands-on lab assignments (process mapping, SPC control chart construction, regression analysis in Excel) and simulation exercises.	30%
Assessed through final project presentations (Quality Dashboard design), EFQM/Baldrige strategic integration projects, and term examinations.	25%

PEDAGOGY:

- **Professor-led Lectures & Discussions:** Introduce theoretical foundations, historical evolution, quality philosophies (Deming, Juran, Crosby), and organizational behavior models.
- **Case Studies & Case Workshops:** Real-world case analyses (e.g., Toyota Harvard Case, SERVQUAL, QFD) to ground quality management in live business contexts.
- **Hands-on Labs & Software Tools:** Practical sessions applying statistical quality tools (SPC, Pareto, Run Charts, Regression) and building executive dashboards in Power BI/Excel.
- **Simulations & Group Activities:** Interactive workshops for root cause analysis (5 Whys, Fishbone), process/value stream mapping, FMEA execution, and group quality culture activities.
- **Guest Lectures & Seminars:** Expert-led sessions focusing on Industry 4.0, IoT, AI integration, smart manufacturing, and contemporary digital transformation.
- **Capstone Project Presentations:** Group projects integrating business excellence frameworks (EFQM/Baldrige) into a unified quality strategy.

SESSION PLAN:

20 sessions of 90 minutes of each, totalling 30 Hours for this 3-credit course.

Session Plan			Pedagogy
Sl. No.	**Topic**	**Pre-reading / Reference**	
1	**Module 1 – Foundations of Quality Leadership** Evolution of Quality: From Inspection to TQM	Deming, Juran, Crosby	Lecture, Case
2	Quality Philosophies and Leadership	Harvard Case on Toyota	Discussion
3	Quality Culture and Organizational Behavior	"Quality is Free" – Crosby	Group Activity
4	**Module 2 – Diagnosing and Measuring Quality** Voice of the Customer (VOC) and Kano Model	Parasuraman SERVQUAL	Case, Workshop
5	Measurement Systems & KPIs	ISO 9001 Guidelines	Exercise
6	Cost of Quality (COQ) and Defect Metrics	-	Lecture, Problem Solving
7	**Module 3 – Improvement Frameworks and Toolkit** PDCA, DMAIC, and Kaizen Methodologies	Imai, "Kaizen"	Workshop
8	Root Cause Analysis (5 Whys, Fishbone)	5 Whys vs Fishbone Diagram	Simulation
9	Process Mapping & Value Stream Mapping	-	Lab
10	Failure Mode and Effects Analysis (FMEA)	-	Activity
MID-TERM EXAMINATION			
11	**Module 4 – Advanced Analytical Tools** Statistical Process Control (SPC) & Control Charts	Montgomery, "Quality Control"	Lecture, Excel

Session Plan			Pedagogy
Sl. No.	**Topic**	**Pre-reading / Reference**	
12	Pareto, Histogram, and Run Charts	-	Practical
13	Regression and Correlation for Quality	-	Lab
14	**Module 5 – Strategic & Organizational Quality** Quality Function Deployment (QFD)	What is Quality Function Deployment (QFD)? By ASQ	Case Study
15	Six Sigma, Lean Thinking & Benchmarking	-	Workshop
16	Module 6 – Digital Quality & Future Excellence Change Management & Employee Involvement	Kotter Model	Discussion
17	Digital Transformation in Quality (IoT, AI, Analytics)	-	Seminar
18	Smart Manufacturing and Industry 4.0	McKinsey Reports	Guest Lecture
19	Designing a Quality Dashboard	Power BI / Excel	Lab
20	Integrating Quality Strategy with Business Excellence	EFQM / Baldrige Framework	Project Presentation
END-TERM EXAMINATION			

RECOMMENDED TEXTBOOK:

Montgomery, D. C. (2019). *Statistical Quality Control* (8th ed.). John Wiley & Sons

REFERENCE TEXTBOOKS/RESOURCES:

- Crosby, P. B. (1979). *Quality is Free: The Art of Making Quality Certain*. McGraw-Hill.
- Imai, M. (1986). *Kaizen: The Key to Japan's Competitive Success*. McGraw-Hill.
- Gryna, F. M., Chua, R. C. H., & Defeo, J. A. (2007). *Juran's Quality Planning and Analysis for Enterprise Quality* (5th ed.). McGraw-Hill.
- McKinsey & Company Reports on Smart Manufacturing and Digital Quality.
- EFQM Excellence Model / Malcolm Baldrige National Quality Award Framework Documentation.

COURSE EVALUATION:

Evaluation is a continuous process at SOIM. Students will be evaluated through continuous assessments, mid-term examinations, and end-term projects/examinations. The evaluation scheme is as follows:

Evaluation	Weightage
(I) Continuous Evaluation: (Quizzes: 10%; Assignments and Group/Individual Projects: 30%)	40%
(II) Mid Term Examination	20%
(III) Final Examination	40%
Total	**100 %**

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